The Geometry of Moral Representation: Framework-Specific Encoding and Inter-Framework Structure in Language Models

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Abstract

How do language models organize moral knowledge internally? We move beyond binary moral detection (moral vs. neutral) to structured moral representation by training six independent linear probes—one per Moral Foundations Theory (MFT) foundation—at each layer of OLMo-2 1B and OLMoE-1B-7B. The learned probe weight vectors define *directions* in representation space whose angular relationships reveal how the model organizes moral concepts. We find the *integration* signature: foundation directions are distinct (effective dimensionality = 5 of 6) and share a common moral-salience component (mean pairwise cosine similarity $\approx$0.22–0.27 across layers), but do not cluster into MFT-predicted individualizing/binding groups. The model develops genuine multi-dimensional moral structure that is not aligned with human moral-psychological categories. This geometry is consistent across dense and MoE architectures and stabilizes within the first 2,000 training steps—well before probing accuracy saturates. Framework-specific fragility testing reveals that output dilution is not foundation-uniform: sanctity/degradation is the most robust foundation in the dense model but the most fragile in MoE (6.2$\times$ ratio), far exceeding the overall architectural gap. Extending to moral dilemmas, we find partial compositionality: dilemma probe directions share $\sim$10% of their variance with the subspace spanned by their component foundations (100$\times$ the null baseline), with near-balanced loading and a complexity–fragility gradient, yet $\sim$90% residual encoding conflict-specific structure.

1 Introduction

How do language models represent morality? Prior work in this series established that models encode moral content *broadly*: probing accuracy saturates early during pre-training and spans nearly all layers (Reblitz-Richardson, 2026b). Fragility testing revealed that this encoding grows more robust throughout training even after accuracy plateaus. Extension to mixture-of-experts (MoE) models showed that moral signal is uniform across experts, with no expert specialization, but that a 74$\times$ output scale gap produces structural fragility (Reblitz-Richardson, 2026a).

Both papers treated moral encoding as a single binary feature: moral vs. neutral. A model that merely detects "this text involves morality" has cleared a low bar. Genuine moral understanding requires *structured* representations: the ability to distinguish care from fairness, loyalty from authority, and to encode the relationships between them. The transition from moral *detection* to moral *understanding* is the subject of this paper.

We operationalize this transition through the geometry of foundation-specific probe directions. Where prior work trained one probe to separate moral from neutral content, we train six: one for each Moral

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Foundations Theory (MFT) foundation (Graham et al., 2013, Haidt, 2012). The learned probe weight vectors define *directions* in the model's representation space. The angular relationships between these directions reveal whether the model has developed structured moral representations.

Three geometric signatures correspond to three qualitatively different modes of moral representation:

1. **Collapse.** All foundation directions converge toward a single "moral salience" direction. The model detects moral relevance but does not distinguish frameworks. This is detection without structure.
2. **Isolation.** Foundation directions are orthogonal with no relational structure. The model has separate moral "slots" but no representation of how frameworks relate. This is structure without coherence.
3. **Integration.** Foundation directions are separated but non-orthogonal, with inter-framework geometry reflecting known relationships from moral psychology. This is the precondition for moral reasoning.

Applied to OLMo-2 1B and OLMoE-1B-7B, we find clear *integration*: foundation directions are distinct (mean pairwise cosine similarity ≈ 0.22–0.27 across layers, far from collapse) and span 5 effective dimensions at every layer. However, hierarchical clustering does not recover the MFT distinction between individualizing and binding foundations (the inter-framework structure the model develops is not aligned with human moral-psychological categories, despite the directions being well-separated).

Extending the fragility protocol to per-foundation probes shows differential robustness: sanctity/degradation is the most robust foundation in dense models but the most fragile in MoE, showing a $6.2\times$ fragility ratio that far exceeds the overall architectural gap. Output dilution does not suppress all moral foundations uniformly; culturally specific moral concepts appear more vulnerable to the MoE aggregation bottleneck.

This paper makes three contributions:

1. **Framework-specific probing methodology.** We introduce probe direction geometry as a tool for measuring structured moral representations, bridging the gap between binary moral probing and the richer structure posited by moral psychology.
2. **Empirical geometric characterization.** We show that OLMo-2 1B and OLMoE-1B-7B exhibit the integration signature, that framework geometry is comparable across architectures, and that the model's inter-framework structure does not align with MFT group predictions despite clear directional separation.
3. **Differential fragility.** We demonstrate that moral foundations differ in robustness, and that MoE architecture disproportionately degrades sanctity/degradation representations, the first evidence that output dilution is not foundation-uniform.

2 Related Work

2.1 Probing for linguistic and semantic structure

Linear probing (Alain and Bengio, 2017) has become a standard tool for reading off what information is encoded in neural network representations. Conneau et al. (2018) probed sentence embeddings for syntactic properties; subsequent work probed for part-of-speech, dependency relations, coreference, and world knowledge. The linear representation hypothesis (Park et al., 2024) formalizes the claim that concepts are encoded as directions in representation space, precisely the assumption underlying our use of probe weight vectors as geometric objects.

Two limitations of the probing literature motivate our approach. First, most probing studies report only *accuracy*, discarding the learned probe parameters. We show that the probe weight vector (the normal to the classification hyperplane) carries geometric information about how concepts relate to each other. Second, probing studies typically train one probe per concept, treating concepts as independent. Our multi-probe geometric analysis recovers inter-concept structure from independently trained probes.

2.2 Geometry of concept representations

Bolukbasi et al. (2016) demonstrated that gender bias in word embeddings manifests as a geometric subspace, and that debiasing amounts to projecting out a direction. Arditi et al. (2024) showed that refusal behavior in instruction-tuned LLMs is mediated by a single direction in activation space. These results establish the precedent that high-level behavioral properties can be localized to specific directions.

Our work extends this line from single directions to *sets of related directions*. Where prior work asked "where is concept X?" we ask "what is the geometric relationship between concepts $X_1, \ldots, X_k$?" The cosine similarity matrix between foundation probe directions is a form of representational similarity analysis (RSA; Kriegeskorte et al., 2008) applied not to stimulus response patterns but to the probes that decode them.

2.3 Moral psychology and Moral Foundations Theory

Moral Foundations Theory (MFT; Graham et al., 2013, Haidt, 2012) posits that human moral judgment draws on (at least) five or six innate foundations: care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation, and (later added) liberty/oppression. MFT predicts a structural distinction between *individualizing* foundations (care, fairness, liberty), which protect individuals from harm, and *binding* foundations (loyalty, authority, sanctity), which bind individuals into groups. This distinction has been validated in cross-cultural surveys and predicts political orientation (Graham et al., 2013).

Alternative taxonomies exist. Curry et al. (2019) propose morality-as-cooperation, identifying seven moral domains grounded in evolutionary game theory. Our use of MFT is pragmatic: the six-foundation taxonomy gives a tractable set of directions for geometric analysis, and the individualizing/binding prediction gives a testable structural hypothesis.

2.4 Moral reasoning in language models

Hendrycks et al. (2021) introduced the ETHICS benchmark for evaluating moral reasoning in language models across justice, deontology, virtue, utilitarianism, and commonsense dimensions. Most work in this area evaluates models *behaviorally*, measuring what outputs models produce in response to moral scenarios. Our approach is *representational*: we probe the internal geometry of moral encoding, asking not whether the model produces the right moral judgment but whether it has developed structured representations of moral distinctions.

2.5 Companion papers

This paper is the third in a series. Paper 1 (Reblitz-Richardson, 2026b) established that moral content is decodable by linear probes from the earliest layer of OLMo-2, that probing accuracy saturates early during pre-training, and that fragility testing (injecting Gaussian noise into activations) resolves the continued development of moral encoding after accuracy plateaus. Paper 2 (Reblitz-Richardson, 2026a) extended the analysis to MoE models (OLMoE-1B-7B), finding uniform moral encoding across experts but a 74× output scale gap that produces structural fragility.

The present paper moves from *binary* moral encoding (moral vs. neutral) to *structured* moral encoding (framework-specific directions and their inter-framework geometry). It reuses the probing and fragility protocols from Papers 1 and 2, applying them at the per-foundation level.

3 Methodology

We train six foundation-specific linear probes at each transformer layer, extract the learned weight vectors as geometric directions in representation space, and analyze the angular relationships between these directions. The methodology decomposes into five components: foundation-specific probing (Section 3.2), geometric analysis of probe directions (Section 3.3), bootstrap direction stability assessment (Section 3.4), framework-specific fragility testing (Section 3.5), and the probing dataset (Section 3.6). All experiments run on a single MacBook Pro M4 Pro (24 GB unified memory, MPS backend).

3.1 Models and comparison design

We evaluate two base (non-instruct) models from the same lab (Ai2), matched in layer count (16) but differing in architecture:

- **OLMo-2 1B** (`allenai/OLMo-2-0425-1B`): dense transformer, 1.5B parameters, 2048 hidden dimension (OLMo Team, 2025). Ai2 publishes 37 early-training checkpoints at 1K-step intervals (steps 0–36K), enabling trajectory analysis.
- **OLMoE-1B-7B** (`allenai/OLMoE-1B-7B-0924`): mixture-of-experts, 6.9B total parameters, 1.3B active, 64 experts per layer, top-8 routing, 2048 hidden dimension (Muennighoff et al., 2024).

Both models use the same tokenizer and comparable training corpora. The comparison tests whether framework geometry is an artifact of dense connectivity or a general property of transformer-based language modeling. Reblitz-Richardson (2026a) established that moral encoding in OLMoE is uniform across experts with a 74$\times$ output scale gap relative to OLMo-2; the present work asks whether this output dilution affects the *structure* of moral representations, not just their *scale*.

3.2 Foundation-specific probing

For each of the six Moral Foundations Theory (MFT) foundations (care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation, and liberty/oppression; (Graham et al., 2013, Haidt, 2012)), we train a binary linear probe at each of 16 transformer layers.

Probe architecture. Each probe is `nn.Linear(2048, 1)` trained with BCE loss and Adam (lr = 10^{-2}) for 50 epochs. This matches the probe specification from Reblitz-Richardson (2026b) and Reblitz-Richardson (2026a), enabling direct comparison with their binary moral/neutral probes.

Activation collection. For each text, we run a single forward pass capturing all 16 layers simultaneously. At each layer, we mean-pool across the sequence dimension to obtain a single $\mathbb{R}^{2048}$ vector per text. Positive examples are the foundation-tagged moral sentences; negative examples are their matched neutral counterparts.

Direction extraction. After training, we extract the weight vector $\mathbf{w} \in \mathbb{R}^{2048}$ from each probe and normalize to unit length: $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. This unit vector is the normal to the classification hyperplane, the *direction* in representation space that maximally separates the foundation's moral content from neutral content. We call $\hat{\mathbf{w}}$ the foundation's **probe direction** at a given layer.

This yields $6 \times 16 = 96$ unit-norm probe direction vectors per model.

3.3 Geometric analysis

The paper's core contribution is analyzing the angular relationships between the six foundation probe directions at each layer.

Cosine similarity matrices. At each layer, we compute the 6×6 pairwise cosine similarity matrix of the foundation probe directions. Because the directions are unit-normalized, $\cos(\hat{\mathbf{w}}_i, \hat{\mathbf{w}}_j) = \hat{\mathbf{w}}_i \cdot \hat{\mathbf{w}}_j$.

Geometric signatures. Three qualitative modes of moral representation correspond to distinct cosine similarity patterns:

1. *Collapse (averaging).* All foundation directions converge toward a single "moral salience" direction. Mean pairwise cosine similarity $\to 1$.
2. *Isolation.* Foundation directions are orthogonal with no relational structure. Mean pairwise cosine similarity $\to 0$.
3. *Integration.* Foundation directions are separated but non-orthogonal, with inter-framework geometry reflecting known relationships. Mean pairwise cosine similarity in $(0, 1)$ with structured variation.

Effective dimensionality. We compute the effective dimensionality of the 6-direction set at each layer via PCA on the 6×2048 matrix of probe directions. Effective dimensionality is the number of principal components explaining $\geq 90\%$ of variance. Low dimensionality (1–2) indicates collapse; high dimensionality (5–6) indicates separation.

Hierarchical clustering. We apply Ward's method to the cosine distance matrix $(1 - \cos(\cdot, \cdot))$ at each layer. Dendrograms reveal whether the six foundations cluster into interpretable groups.

Permutation test for MFT group structure. MFT predicts that the six foundations divide into *individualizing* (care, fairness, liberty) and *binding* (loyalty, authority, sanctity) clusters (Graham et al., 2013). We test this prediction with a permutation test: compute the observed difference between mean within-group cosine similarity and mean between-group cosine similarity, then permute group assignments 10,000 times to generate the null distribution.

3.4 Bootstrap direction stability

With 32 training pairs per foundation in 2048 dimensions, the probe has 2049 parameters and only 64 training examples. The classification *accuracy* may be robust in this regime (a single hyperplane suffices for binary separation), but the extracted *direction* could be noisy , and noise in directions contaminates the angular analysis.

We assess direction stability via bootstrap resampling: for each foundation at each layer, we resample the 32 training pairs with replacement 200 times, retrain the probe on each bootstrap sample, and compute the cosine similarity of each bootstrap direction with the full-data direction. A mean bootstrap cosine similarity > 0.8 indicates a stable direction.

This gives a per-layer, per-foundation reliability metric for the geometric analysis. Layers where directions are unstable should be interpreted with caution.

3.5 Framework-specific fragility

We extend the fragility protocol of Reblitz-Richardson (2026b) to per-foundation probes. For each foundation at each layer:

1. Train a linear probe on clean activations.
2. Evaluate on clean test activations (baseline accuracy).
3. For each noise level $\sigma \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$, add $\mathcal{N}(0, \sigma^2)$ noise to cached test activations and re-evaluate.
4. The **critical noise** σ^* is the smallest σ where accuracy drops below 0.6.

This tests the universality hypothesis: more cross-culturally universal foundations (care/harm) should be more robustly encoded than culturally variable foundations (sanctity/degradation, loyalty/betrayal). Cross-architectural comparison tests whether the output dilution effect (Reblitz-Richardson, 2026a) affects all foundations uniformly.

3.6 Probing dataset

We use the same 240-pair minimal-pair probing dataset from Reblitz-Richardson (2026b), now used at the per-foundation level: 40 pairs per MFT foundation, stratified 80/20 into 32 train and 8 test pairs per foundation. Each pair consists of a moral sentence tagged with its MFT foundation and a matched neutral sentence controlling for sentence length, syntactic structure, and topic. The dataset was drawn from a $1\{,\}$200-pair candidate pool generated by Claude Sonnet 4.6 from hand-written seed examples, with automated validation gates (length ratio ≤ 1.5, embedding similarity, keyword scan, deduplication) and LLM-as-judge filtering for neutral-pair quality.

For the geometric analysis, the 32 training pairs per foundation yield 64 training examples (moral + neutral) for the `nn.Linear(2048, 1)` probe. While this is a small sample for a 2049-parameter model, the bootstrap stability analysis (Section 3.4) directly quantifies whether the resulting directions are reliable.

3.7 Dilemma compositionality analysis

Experiments 1–7 establish that the model maintains distinct directions for each moral foundation. A natural follow-up: when two foundations *conflict* in a moral dilemma, does the model represent the dilemma as a composition of its component foundation directions, or does it develop a qualitatively new representation?

Dilemma dataset. We generate 300 moral dilemma scenarios (20 per each of the $\binom{6}{2} = 15$ foundation pairs) using Claude Sonnet 4.6, with hand-written seed examples per pair. Each scenario pits two specific foundations against each other (e.g., care vs. authority: "The nurse administered an unapproved painkiller to a dying patient because following protocol meant hours more agony"). Each dilemma text is paired with a matched neutral sentence. All pairs pass automated validation gates (length ratio, keyword scan, deduplication).

Dilemma-specific probes. For each of the 15 foundation pairs, we train a binary linear probe (same architecture as Section 3.2) to distinguish dilemma moral text from matched neutral text. The probe weight vector $\hat{\mathbf{w}}_{\text{dilemma}}$ is the direction in representation space that separates the dilemma content from neutral content.

Subspace membership score. To measure compositionality, we project each dilemma direction onto the 2D subspace spanned by its two component foundation directions. Given component directions $\hat{\mathbf{w}}_A$ and $\hat{\mathbf{w}}_B$ from Experiment 1, we orthogonalize them via Gram–Schmidt and compute the fraction of the dilemma direction's variance explained by this 2D subspace:

$$S = \|\text{proj}_{\text{span}(\hat{\mathbf{w}}_A, \hat{\mathbf{w}}_B)} \hat{\mathbf{w}}_{\text{dilemma}}\|^2$$

A membership score of $S = 1$ indicates full compositionality (the dilemma direction lies entirely within the component subspace); $S = 0$ indicates complete independence. The null baseline for a random unit vector in $\mathbb{R}^{2048}$ projected onto a random 2D subspace has expected membership $2/2048 \approx 0.001$. We estimate the empirical null distribution from 10{,}000 random unit vectors.

Component balance. We decompose the within-subspace projection into components along $\hat{\mathbf{w}}_A$ and the Gram–Schmidt-orthogonalized $\hat{\mathbf{w}}_B$. The balance ratio (fraction of the projection along the first component) measures whether the dilemma direction is dominated by one foundation or draws equally on both. A ratio near 0.5 indicates balanced composition.

Shared-component geometry. If dilemma representations are partially compositional, dilemma pairs that share a foundation component should have more similar probe directions than pairs with no shared foundation. We test this by comparing the mean cosine similarity between dilemma directions for pairs that share a component (e.g., care–fairness and care–loyalty, which share care) versus pairs with no overlap (e.g., care–fairness and loyalty–sanctity).

Cross-architecture consistency. We repeat the dilemma probing and subspace analysis on OLMoE-1B-7B to test whether the compositionality structure is architecture-specific or general.

4 Results

4.1 Foundation-specific probe accuracy

All six foundation-specific probes achieve perfect or near-perfect accuracy across the full depth of OLMo-2 1B. Every foundation reaches 100% peak accuracy. Authority/subversion achieves 100% at all 16 layers; the remaining foundations show minor fluctuations at individual layers (minimum 87.5% for care/harm at layer 1). The onset threshold (0.6) is exceeded at layer 0 for all foundations.

These results confirm that moral foundation content is linearly separable from the earliest layer, consistent with the "immediate onset" finding of Reblitz-Richardson (2026b) for the pooled binary moral/neutral probe. The per-foundation decomposition shows no qualitative difference in *detectability* across foundations: all are equally easy to decode. The interesting variation is not in accuracy but in the *geometry* of the probe directions that achieve this accuracy.

On OLMoE-1B-7B, all foundations similarly reach 100% peak accuracy, with the lowest early-layer values at 81.25% (layers 0–1). The accuracy profiles are essentially indistinguishable between architectures.

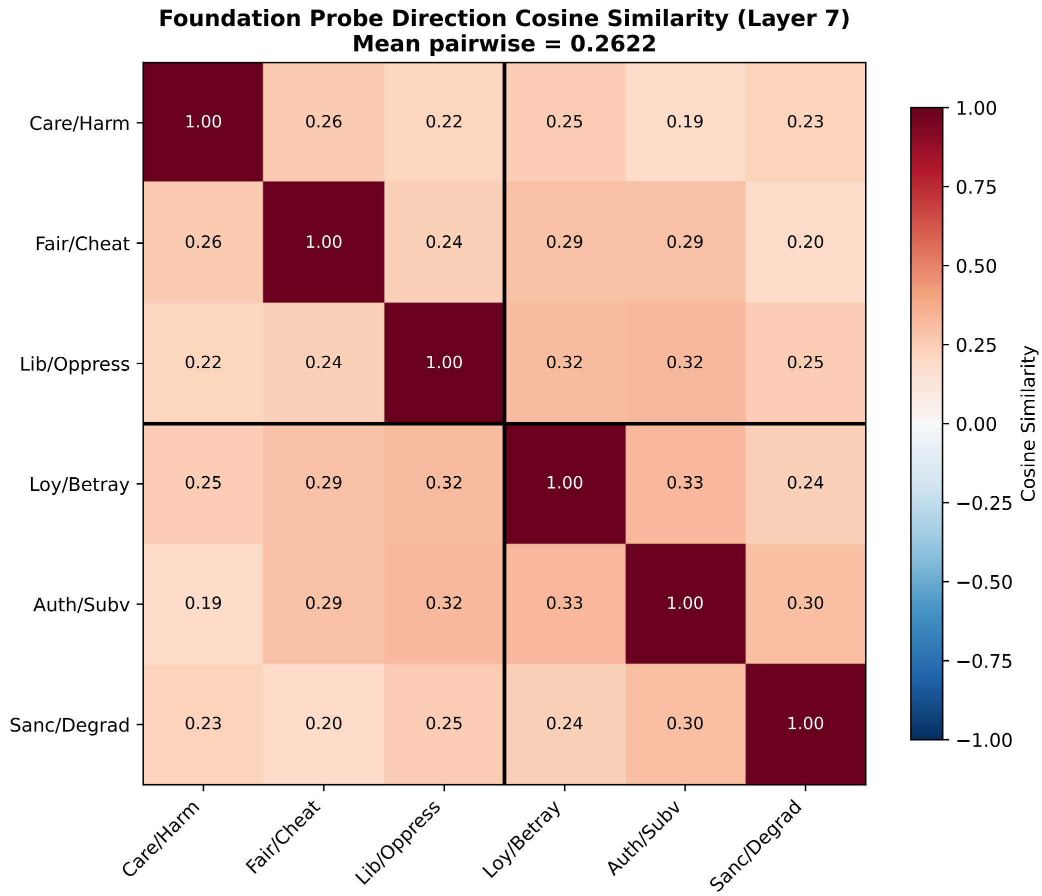

Figure 1: Pairwise cosine similarity between foundation probe directions at layer 7 (bootstrap-stable; all six foundations exceed the 0.8 stability threshold at this layer; Section 4.6), OLMo-2 1B. Mean off-diagonal cosine = 0.262. See Appendix C for matrices at layers 0 and 15.

4.2 Framework geometry: integration, not collapse

The headline finding: foundation probe directions are *separated*, not collapsed. Across bootstrap-stable layers (6–15, where all six directions exceed the 0.8 stability threshold; Section 4.6), mean pairwise cosine similarity ranges from **0.232 to 0.274**, far below the collapse threshold (> 0.95) and below the intermediate zone (0.8–0.95). Figure~1 shows the representative pattern at layer 7 (mean cosine 0.262). The peak-separation layer (layer 0, mean cosine 0.216) is below the bootstrap stability threshold for all foundations and is reported in Appendix~C.

The cosine similarities are uniformly *positive* at all layers (range 0.14–0.35), indicating that the foundation directions share a common component. This is the *integration* signature from our trichotomy: the directions are separated but non-orthogonal, consistent with a shared moral-salience subspace from which foundation-specific directions deviate.

Effective dimensionality. The six foundation directions span 5 effective dimensions (the number of PCs explaining $\geq 90\%$ of variance) at every layer. This is near the maximum possible for 6 directions, confirming that the directions are geometrically distinct; they do not collapse into a lower-dimensional subspace.

4.3 Dendrogram structure does not recover MFT groups

Hierarchical clustering of the six foundation directions does *not* recover the MFT individualizing/binding distinction at any layer. At layer 7, loyalty and authority merge first, liberty joins them,

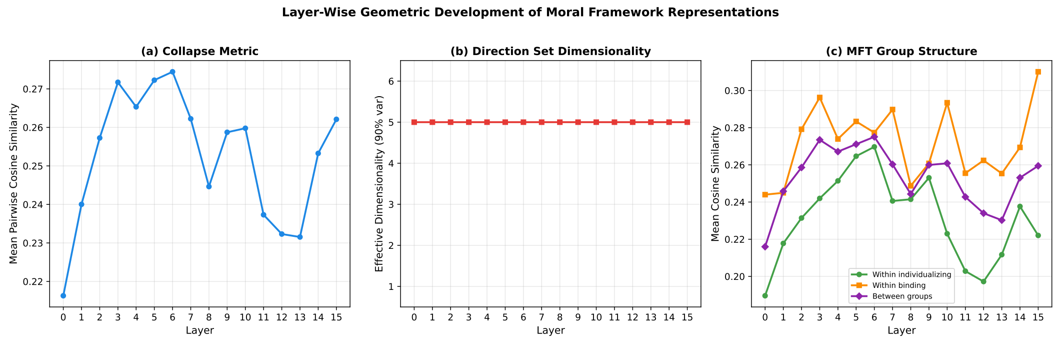

Figure 2: Layer-wise geometric metrics for OLMo-2 1B. (a) Mean pairwise cosine similarity is relatively flat across layers. (b) Effective dimensionality remains constant at 5 across all layers.

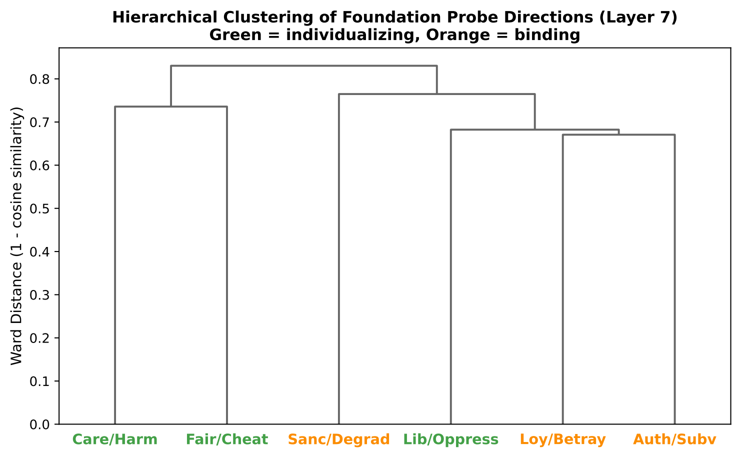

Figure 3: Hierarchical clustering (Ward's method) of foundation probe directions at layer 7 (bootstrap-stable), OLMo-2 1B. The first split does not recover the MFT individualizing/binding distinction.

then care and fairness merge separately, then sanctity joins the loyalty–authority–liberty cluster. No layer produces the predicted {care, fairness, liberty} vs. {loyalty, authority, sanctity} partition.

The most consistent clustering pattern across layers is a care–sanctity pairing: these two foundations appear in the same cluster at 10 of 16 layers. This crosses the MFT boundary (care is individualizing, sanctity is binding) but has a plausible semantic interpretation: both foundations concern protection of vulnerable entities (persons from harm, sacred things from degradation). The loyalty–authority pairing at layer 7 is within the binding group, but liberty (individualizing) joins this cluster before grouping with the other individualizing foundations (care, fairness), further mixing MFT categories.

The permutation test for the individualizing/binding distinction does not reach significance at any layer (minimum $p = 0.32$; median $p = 0.53$). With only 6 items and 20 unique 3–3 partitions, statistical power is limited, but the consistently high p-values combined with the absence of MFT-aligned dendrograms at any layer indicate that the model's inter-framework geometry does not reflect the MFT group structure on this dataset.

4.4 Layer-wise geometric development

The geometric structure is relatively stable across layers. Mean pairwise cosine similarity is lowest at layer 0 (0.216), rises modestly to a peak of 0.274 at layer 6, then returns to 0.232 at layer 13 before a slight increase at layer 15 (0.262). The variation is small (range 0.06) compared to the distance from collapse, indicating that framework separation is a consistent property of the representation space rather than an emergent late-layer phenomenon.

Effective dimensionality remains constant at 5 across all layers, meaning the rank of the direction set does not change even as the pairwise angles shift. The geometric structure is a rotation of the direction set within a fixed-rank subspace, not a dimensionality change.

4.5 Dense vs. MoE framework geometry

Framework geometry is remarkably similar between architectures. OLMoE-1B-7B shows mean pairwise cosine similarity ranging from 0.219 to 0.287, compared to OLMo-2 1B's 0.217 to 0.282. Effective dimensionality is 5 at all layers for both models. The overall degree of framework separation is consistent across dense and MoE architectures in our comparison.

Neither architecture produces MFT-aligned dendrogram structure at any layer. Both models show the care–sanctity pairing as the most stable clustering feature, suggesting that inter-framework geometry is driven by semantic relationships in the training corpus rather than by architectural properties. The permutation test for MFT group structure is non-significant at all layers for both models (all $p > 0.25$).

This finding extends Reblitz-Richardson (2026a): output dilution affects moral encoding *scale* (74× signal gap) but not framework *structure*. The geometric organization of moral foundations is preserved across architectures.

4.6 Direction stability under bootstrap

Bootstrap resampling (200 iterations) shows a stability gradient across layers. Early layers (0–5) show borderline stability (mean cosine similarity with the full-data direction: 0.74–0.80, below the 0.8 threshold for most foundations). Middle and late layers (6–15) are stable (mean cosine > 0.80). Sanctity/degradation is the most stable foundation (13/16 layers stable); care/harm the least (10/16 stable).

This gradient has two implications. First, the geometric analysis at early layers, including the peak separation layer (layer 0), should be interpreted with caution, as the specific pairwise cosine values may shift under resampling. Second, the stability gradient itself is informative: probe directions become more determined as representations become more specialized, paralleling the lexical-to-compositional gradient from Reblitz-Richardson (2026b).

The headline geometric findings (separation not collapse; effective dimensionality = 5) are confirmed at layers 6–15 where directions are stable.

4.7 Differential fragility across frameworks

Per-foundation fragility shows differential robustness within both architectures.

OLMo-2 1B (dense). Sanctity/degradation is the most robust foundation (mean critical noise $\sigma^* = 5.60$), followed by authority/subversion (5.02), care/harm (4.42), liberty/oppression (3.82), fairness/cheating (3.52), and loyalty/betrayal (3.31). The universality hypothesis (care/harm as most robust) is *not* supported. Binding foundations as a group are slightly more robust (mean 4.64) than individualizing foundations (mean 3.92), but the difference is driven by the two extremes (sanctity and loyalty) rather than a clean group separation.

OLMoE-1B-7B (MoE). The ordering shifts: loyalty/betrayal is most robust (2.03), followed by liberty/oppression (1.60), authority/subversion (1.35), care/harm (1.26), fairness/cheating (1.01), and sanctity/degradation (0.91). Binding foundations remain slightly more robust as a group (mean 1.43 vs. 1.29 for individualizing), preserving the direction of the dense-model difference.

The most striking finding is per-foundation: sanctity/degradation is the *most* robust foundation in the dense model (5.60) but the *least* robust in MoE (0.91). This 6.2× ratio is far larger than the overall per-foundation fragility gap between architectures (3.1×, computed as the ratio of mean critical noise across six per-foundation probes with 32 training pairs each; the 5.1× gap reported by Reblitz-Richardson (2026a) uses a single pooled binary probe with 192 training pairs, which has higher statistical power). Output dilution does not suppress all moral foundations uniformly; sanctity representations are disproportionately vulnerable to the MoE aggregation bottleneck. This may reflect the encoding mechanism: sanctity/purity concepts, which rely on culturally specific associations

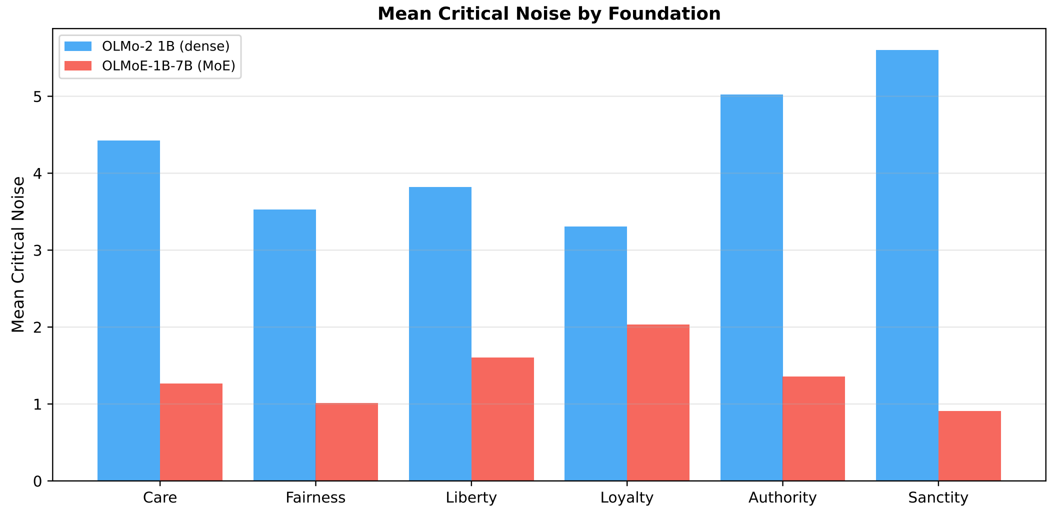

Figure 4: Mean critical noise per foundation for OLMo-2 1B (left) and OLMoE-1B-7B (right). Foundation ordering differs between architectures; sanctity is the most robust in dense and most fragile in MoE.

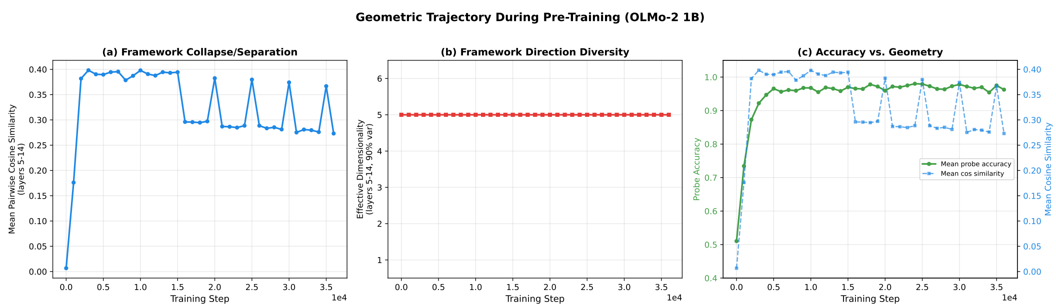

Figure 5: Geometric trajectory during OLMo-2 1B pre-training. (a) Mean cosine similarity stabilizes by step 2000. (b) Effective dimensionality remains at 5 throughout. (c) Accuracy continues climbing after geometry plateaus.

(Graham et al., 2013), may depend on fine-grained signal that is preferentially attenuated by top-k averaging.

4.8 Geometric trajectory during training

Framework geometry develops rapidly and stabilizes early. Tracking mean pairwise cosine similarity (averaged over stable layers 5–14) across 20 OLMo-2 1B checkpoints from step 0 to step 35,000:

- **Step 0.** Mean cosine similarity is 0.007; the six foundation directions are effectively orthogonal, as expected for probes trained on random representations. Mean accuracy is 0.510 (chance).
- **Step 1000.** Cosine similarity jumps to 0.176. Accuracy reaches 0.734. The model has already begun developing shared moral-salience structure.
- **Step 2000.** Cosine similarity reaches 0.382, within 5% of its final value. Accuracy is 0.873, still 10 points below its peak.
- **Steps 3000–15,000.** Cosine similarity plateaus at 0.38–0.40. Accuracy continues climbing from 0.922 to 0.970.
- **Steps 20,000–35,000.** Cosine similarity drifts slightly downward (0.382 to 0.367). Accuracy stabilizes at 0.975–0.979.

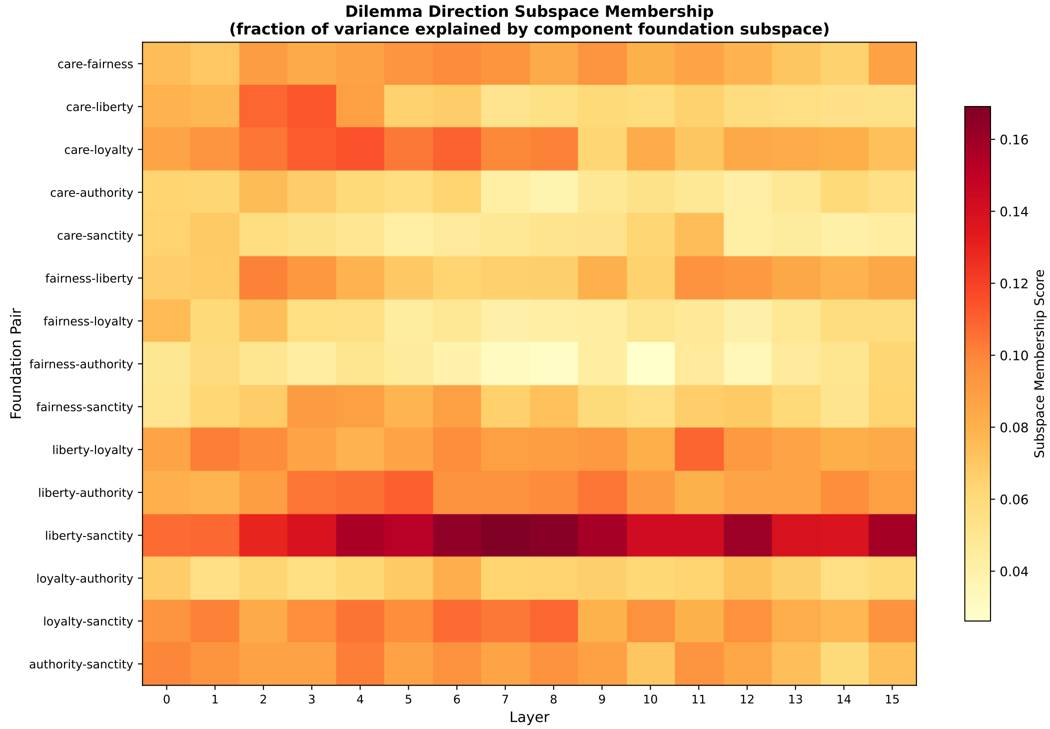

Figure 6: Subspace membership scores for 15 dilemma probe directions across 16 layers. Each cell shows the fraction of a dilemma direction's variance explained by the 2D subspace of its component foundation directions. Liberty–sanctity shows the strongest compositional signal.

Effective dimensionality remains constant at 5 from step 0 onward. This is expected: random unit vectors in $\mathbb{R}^{2048}$ are nearly orthogonal with high probability, so even at initialization the six probe directions span 5 effective dimensions. The informative signal is cosine similarity, which transitions from ≈ 0 (random) to ≈ 0.4 (mature structure) during the first 2000 training steps.

The temporal dissociation (framework geometry stabilizing at step 2000 while accuracy continues improving through step 25,000) extends the two-phase pattern identified by Reblitz-Richardson (2026b): accuracy saturates early, but fragility continues resolving. Here we add a third metric: inter-framework *structure* also stabilizes before inter-framework *discriminability* finishes developing. The model discovers the geometric layout of moral concepts early and then spends the remainder of training strengthening the representations within that fixed layout.

4.9 Dilemma compositionality: partial but structured

We now ask whether the model's representation of moral *dilemmas* (scenarios where two foundations conflict) can be decomposed in terms of the single-foundation directions from Experiment 1.

Dilemma probes achieve high accuracy. All 15 dilemma-specific probes achieve $\geq 75\%$ peak test accuracy (mean 94.2%), with 13 of 15 pairs at $\geq 87.5\%$. The model reliably distinguishes dilemma moral content from matched neutral text.

Subspace membership: partial compositionality. The mean peak subspace membership score across all 15 pairs is **0.099**, approximately 100× the null baseline of 0.001 ($p_{95} = 0.003$; $p_{99} = 0.004$). Every pair exceeds the 99th percentile of the null distribution at its peak layer. However, 0.099 is far from 1.0: on average, only $\sim$10% of each dilemma direction's variance is explained by its component foundation subspace. The remaining $\sim$90% (mean residual norm = 0.949) lies in directions orthogonal to both component foundations.

Component balance is near-equal. The mean component balance ratio is 0.486 (perfect balance = 0.5). In 14 of 15 pairs, the balance falls within [0.40, 0.58], indicating that both component

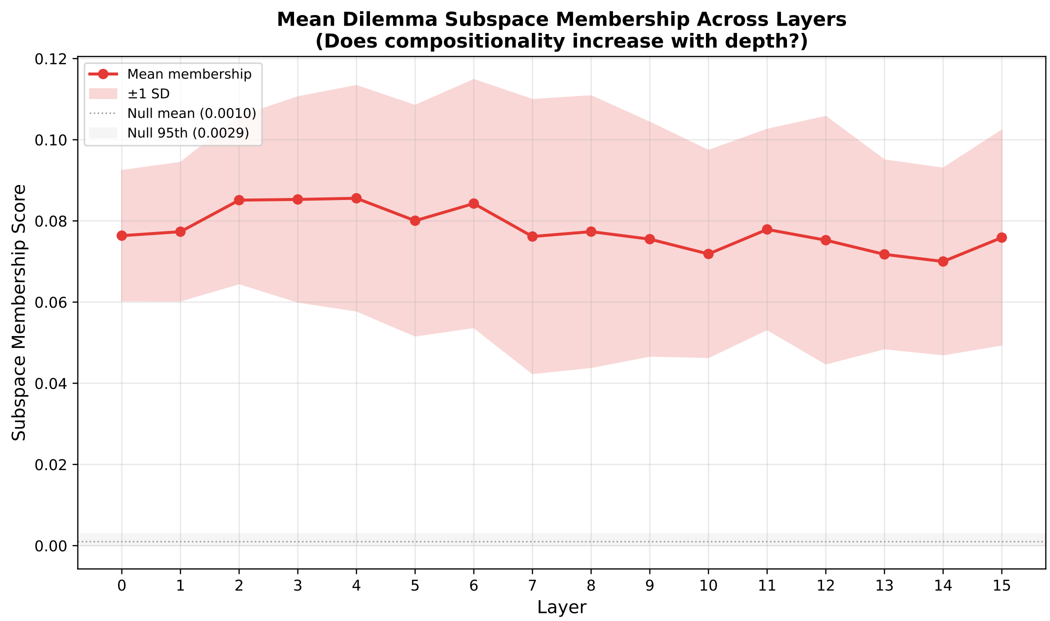

Figure 7: Mean subspace membership across layers (red), with ± 1 SD band. The null baseline (gray) is ~ 0.001. Membership is relatively flat at $\sim 8\%$ across all layers.

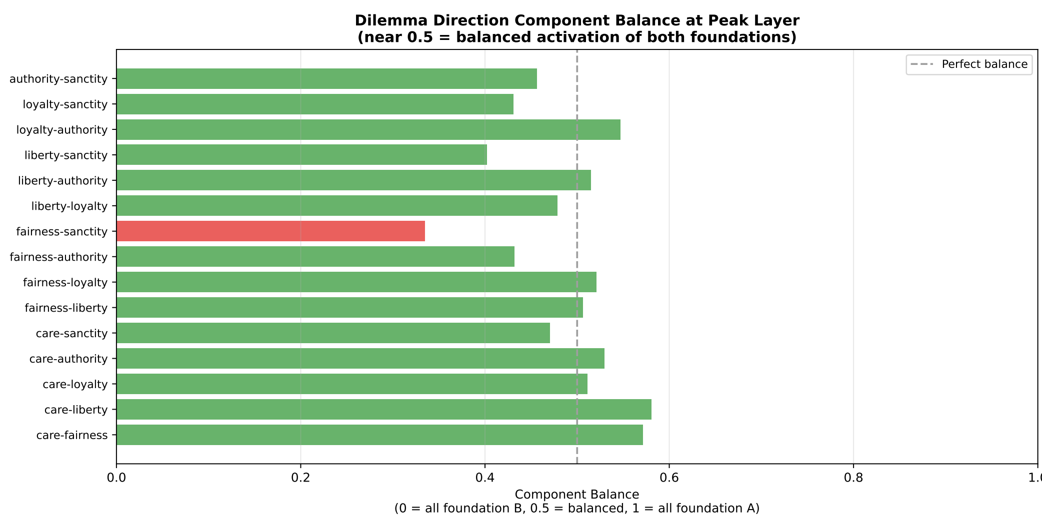

Figure 8: Component balance at each pair's peak subspace membership layer. Values near 0.5 (dashed line) indicate balanced contribution from both foundations. Fairness–sanctity (red) is the only pair with substantial imbalance.

foundations contribute approximately equally to the within-subspace projection. The exception is fairness–sanctity (balance = 0.335), where the sanctity component dominates. When two foundations *do* compose, they compose symmetrically rather than one foundation dominating the representation.

Shared-component structure. Dilemma pairs that share a foundation component have consistently higher cosine similarity than pairs with no shared foundation. At the peak effect layer (layer 13), the mean cosine similarity between shared-component pairs is **0.269** versus **0.195** for non-sharing pairs (difference = 0.074). This difference is positive at every layer, indicating that the compositional structure is not layer-specific but a general property of the dilemma direction geometry.

Hierarchical clustering. The 21-direction dendrogram (6 foundation + 15 dilemma directions) at layer 13 shows two features. First, the six foundation directions form a distinct cluster separate from

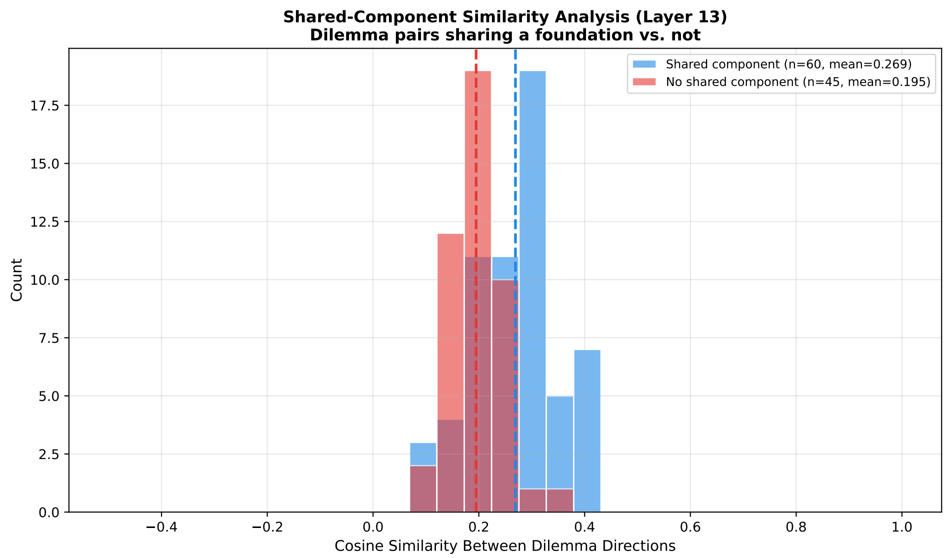

Figure 9: Distribution of pairwise cosine similarities between dilemma directions at layer 13, split by whether the pairs share a component foundation. Shared-component pairs (blue, $n = 60$) have higher mean similarity (0.269) than non-sharing pairs (red, $n = 45$, mean 0.195).

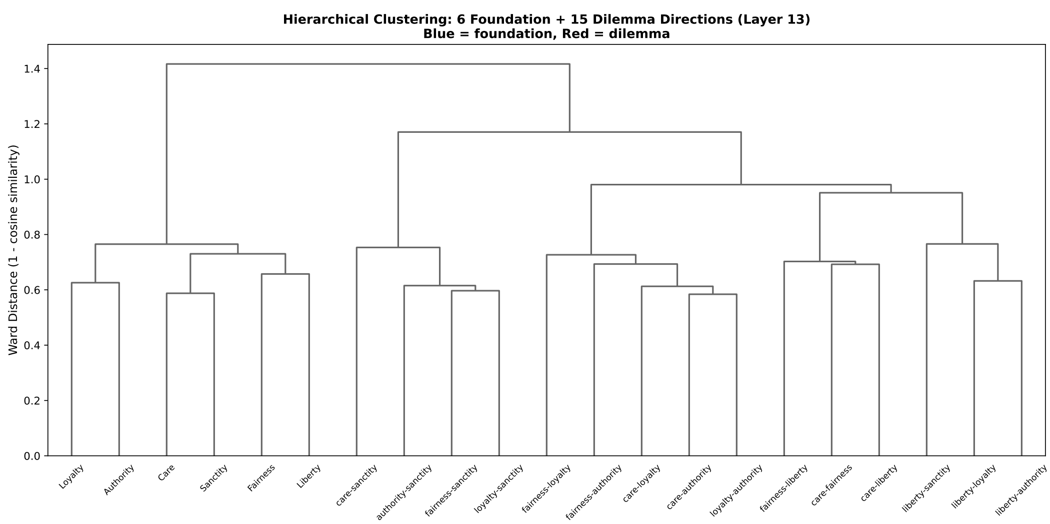

Figure 10: Hierarchical clustering of all 21 directions (6 foundation in blue, 15 dilemma in red) at layer 13. Foundation directions cluster separately from dilemma directions.

the dilemma directions. Second, within the dilemma cluster, pairs sharing a component foundation tend to merge at lower distances, consistent with the shared-component analysis above.

The categorical foundation/dilemma separation initially appears to undercut the compositionality narrative: if dilemmas were purely compositional, they would cluster near their component foundations, not in a separate region. However, this separation is driven by register features, not moral content. Projecting all 21 directions into the 5D moral subspace (spanned by the six foundation directions) and re-clustering dissolves the separation entirely: foundations now cluster with their related dilemmas (e.g., sanctity with fairness–sanctity and care–sanctity). The first-order separation in the full-space dendrogram reflects the $\sim$90% extra-moral residual (likely text-register differences between declarative single-foundation sentences and narrative dilemma scenarios), while the second-order structure within each cluster reflects genuine moral content relationships.

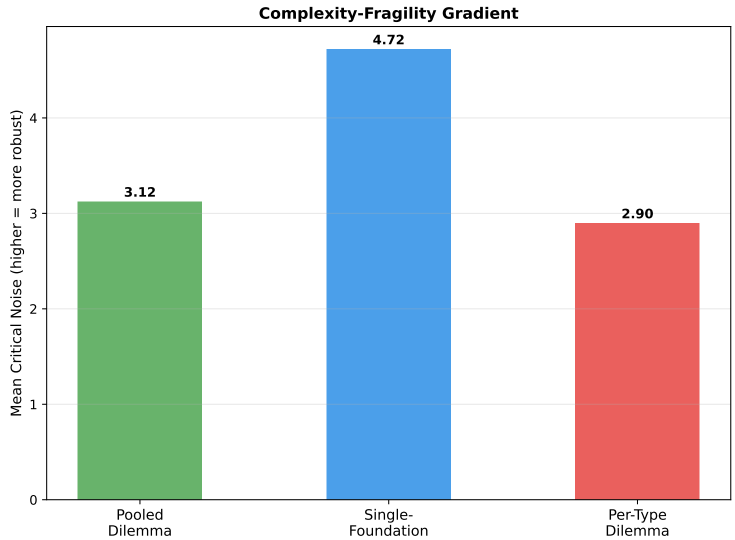

Figure 11: Mean critical noise for probes at three complexity levels. Single-foundation probes (Exp. 7) are most robust; per-type dilemma probes are least robust. Higher values indicate greater noise tolerance.

Full moral subspace projection. The 2D subspace analysis above uses only the two component foundations per dilemma. To test whether dilemma representations are compositional over the *full* moral vocabulary, we project each dilemma direction onto the 5D subspace spanned by all six foundation directions (5D because the six directions have effective dimensionality 5). Averaging across all layers and pairs, the mean 5D membership is **0.093**, only modestly above the layer-averaged 2D membership of 0.078 (ratio 1.19×). (The higher value of 0.099 reported above is the mean of per-pair *peak* values; 0.078 is the cross-layer average, used here for an apples-to-apples comparison with the 5D statistic.) Both exceed their respective null baselines (5D: 38× null; 2D: 81× null), confirming genuine moral content in the dilemma representations. However, the small gain from 2D to 5D indicates that the ∼90% residual is not explained by *any* foundation direction: it is genuinely extra-moral, likely encoding conflict-specific features such as trade-off framing and tension that lie outside the moral subspace entirely.

4.10 Dilemma direction stability

Bootstrap resampling (50 iterations) of the 15 dilemma probe directions across 16 layers yields 240 direction–layer combinations. Of these, **239 are stable** (mean cosine with full-data direction > 0.7), with the single exception at liberty–loyalty, layer 0. The relaxed threshold (0.7 vs. 0.8 for foundation probes) reflects the smaller sample size (16 training pairs vs. 32 for foundations), but the results indicate that the dilemma probe directions, and therefore the subspace analysis built on them, are reliable.

4.11 Complexity–fragility gradient

We compare fragility across three levels of moral complexity: single-foundation probes (from Experiment 7), pooled binary dilemma probes (all 300 dilemma pairs pooled), and per-type dilemma probes (15 separate probes, 20 pairs each).

The mean critical noise follows a **complexity–fragility gradient**: single-foundation probes are most robust ($\sigma^* = 4.72$), followed by the pooled dilemma probe ($\sigma^* = 3.12$), with per-type dilemma probes least robust ($\sigma^* = 2.90$). This ordering is consistent with the hypothesis that more specific moral distinctions are encoded with less redundancy and are therefore more vulnerable to perturbation.

The gradient's direction matters: the pooled dilemma probe is *less* robust than single-foundation probes, not more. This argues against the possibility that dilemma probes are simply detecting a generic "morally complex" feature with high redundancy. Instead, the dilemma direction captures genuinely more specific information that is correspondingly more fragile.

4.12 MoE architecture preserves compositionality

The dilemma probing and subspace analysis on OLMoE-1B-7B produces results consistent with the dense model. Mean peak accuracy is 95.8% (vs. 94.2% on OLMo-2), and mean peak subspace membership is **0.092** (vs. 0.099). The $\sim$7% difference in membership scores is within the variation across individual foundation pairs. The partial compositionality structure (statistically significant but low absolute membership, with high residual) is a property of the representation geometry, not a dense-architecture artifact.

4.13 Robustness to direction-finding method

The geometric findings reported above rely on probe weight vectors as foundation directions. To test whether the geometry is an artifact of discriminative probe training, we extract directions using two alternative methods and compare.

Mean-difference directions. For each foundation, we compute $\mathbf{d}_f = \bar{\mathbf{a}}_{\text{moral}} - \bar{\mathbf{a}}_{\text{neutral}}$ (the normalized difference of class-conditional activation means), requiring no optimization. These training-free directions replicate the core geometric findings: effective dimensionality is 5 at all layers. Like the probe-weight directions, the mean-diff dendrogram does not recover the MFT individualizing/binding split at any layer, and the permutation test is non-significant throughout. Per-foundation cosine similarity between probe-weight and mean-difference directions ranges from 0.67 to 0.72 (mean across layers), indicating related but non-identical directions: the probe-weight method finds more foundation-specific discriminative signal, while the mean-difference method captures more of the shared moral-salience component (mean pairwise cosine 0.41 vs. 0.22 at layer 0).

Representation-engineering directions. We also test paired-difference PCA (Zou et al., 2023): for each pair~i, compute $\mathbf{d}_i = \mathbf{a}_{\text{moral},i} - \mathbf{a}_{\text{neutral},i}$ and take the first principal component. This method performs poorly: the first PC explains only 8–11% of variance (barely above chance in $\mathbb{R}^{2048}$), and the resulting directions show low alignment with probe-weight directions ($|\cos| = 0.07$–0.26) and weak classification accuracy. With $\sim$32 pairs per foundation in 2048 dimensions, the $p \gg n$ regime prevents PCA from isolating the concept direction. This negative result validates that the convergent probe-weight and mean-difference findings are not trivially recoverable; they depend on direction-finding methods with appropriate inductive bias for small datasets.

Cross-register transfer. Both probe-weight and mean-difference directions transfer to narrative dilemma text with $> 90\%$ mean pair accuracy across foundations (the fraction of pairs where the direction projects the moral text higher than the neutral text). Probe-weight directions achieve $> 95\%$ for all foundations; the mean-difference method drops to $\sim$91% for authority/subversion, the foundation with weakest directional stability. The transfer gap between same-register (declarative test pairs) and cross-register (dilemma pairs) averages under 4 percentage points, with authority/subversion showing the largest gap ($\sim$9 pp for mean-difference; §5.5).

5 Discussion

5.1 Integration as the default geometric mode

The central finding, that foundation probe directions exhibit integration rather than collapse or isolation, has a straightforward interpretation: language models develop structured moral representations from distributional statistics alone. The training corpus does not label texts with MFT foundations, yet the model learns representations in which moral foundations occupy distinct directions that share a positive common component (mean cosine ≈ 0.22 at peak separation, ≈ 0.26 at stable mid-network layers). This is genuine multi-dimensional moral structure, not a single "moral salience" detector.

The fact that effective dimensionality is 5 (near-maximal for 6 directions) throughout the network rules out the possibility that the model has learned a single "this text is moral" feature and then adds minor perturbations per foundation. The foundation directions are geometrically distinct objects that happen to share a common moral-salience component.

Dataset sensitivity. The mean cosine similarity is sensitive to neutral-pair quality: neutrals that inadvertently carry moral content inflate the shared moral-salience component, producing higher cosine values. The quality-gated dataset

used here (Section 3.6) minimizes this inflation, and we interpret the \$$\approx$ \$$0.22 - -0.27 rangeasalowerboundonthetrueintegrationsignal.Allqualitativeconclusions(integrationsignature, effectived$
$pairqualityforgeometricanalyses.$

The model's moral taxonomy is not MFT. The inter-framework structure that emerges does not align with MFT's predicted individualizing/binding distinction: hierarchical clustering does not recover this partition at any layer, and the permutation test is non-significant throughout. Instead, the most consistent clustering feature is a care–sanctity pairing that crosses MFT groups. Both care and sanctity involve protection (of persons from harm, of sacred things from degradation), sharing distributional signatures that the model detects. The care–sanctity pairing is itself robust to dataset construction choices: it persists across different neutral-pair generation methods and quality thresholds, arguing against a dataset artifact explanation. The model's moral taxonomy is thus empirically grounded in corpus statistics rather than aligned with any a priori grouping from moral psychology, which is itself evidence of genuine structure learning (the model discovers which moral concepts are distributionally related) rather than surface keyword matching.

5.2 The sanctity fragility anomaly

The most unexpected per-foundation finding is the dramatic architecture-dependent behavior of sanctity/degradation: it is the *most* robust foundation in the dense model ($\sigma^* = 5.60$, highest of all six) but the *least* robust in MoE ($\sigma^* = 0.91$, lowest of all six). This 6.2$\times$ ratio far exceeds the overall architectural gap (3.1$\times$ mean across foundations).

One interpretation: sanctity/purity concepts are encoded through culturally specific associations (disgust, bodily integrity, spiritual elevation; (Graham et al., 2013)) that rely on fine-grained distributional features. In a dense model, these features benefit from the full feedforward scale and are robust. Under MoE top-k averaging, the fine-grained signal is disproportionately attenuated because it requires coordinated contribution across multiple expert outputs, which the averaging operation dilutes more than simpler features.

An alternative interpretation is that the training corpus has rich sanctity-related content (religious text, purity discussions) that produces strong dense representations, but this content routes unevenly across experts in MoE, creating inconsistent per-expert sanctity features that average destructively.

At the group level, both architectures show binding foundations slightly more robust than individualizing (dense: 4.64 vs. 3.92; MoE: 1.43 vs. 1.29), but this group-level pattern is small and driven by individual foundation outliers rather than a clean categorical distinction.

5.3 Framework geometry stabilizes before accuracy saturates

The trajectory analysis shows a temporal dissociation: framework geometry (mean cosine similarity between foundation directions) reaches its mature value by approximately step 2000–3000, while probing accuracy continues climbing through step 10000 and beyond. This extends the "accuracy saturates but fragility resolves" finding of Reblitz-Richardson (2026b) to a third metric: the *structure* of moral representations stabilizes before the *strength* of moral representations finishes developing.

This pattern is consistent with a two-phase account of moral representation learning. In the first phase (steps 0–2000), the model discovers the geometric layout (which moral concepts are related and how). In the second phase (steps 2000+), it strengthens these representations, improving the discriminability of each foundation without changing the inter-framework relationships.

Effective dimensionality is 5 from step 0 onward, indicating that even random initialization produces probe directions that span 5 dimensions. This is expected: random unit vectors in $\mathbb{R}^{2048}$ are nearly orthogonal with high probability. The informative signal is not dimensionality but cosine similarity, which jumps from ≈ 0 (random) to ≈ 0.38 (mature structure) during the first 2000 training steps.

5.4 Partial compositionality of moral dilemmas

When two moral foundations conflict in a dilemma scenario, the model's representation is partially compositional: the dilemma probe direction shares statistically significant overlap with the 2D subspace spanned by its component foundation directions (mean membership $S = 0.099$, 100$\times$ the null baseline of 0.001), yet $\sim$90% of the dilemma direction lies outside this subspace.

This partial compositionality has two complementary interpretations. First, the model recognizes moral dilemmas as involving their component foundations (the $\sim$10% subspace membership is not an artifact), as confirmed by permutation testing and by the shared-component structure (dilemma pairs that share a foundation have higher cosine similarity, $\Delta = 0.074$ at layer 13). Second, the model represents something beyond the sum of parts: the $\sim$90% residual captures conflict-specific features (tension, trade-off framing, or contextual modulation) that single-foundation probes do not isolate.

The near-balanced component loading ($\bar{\alpha} = 0.486$) is notable. If dilemma representations were dominated by one foundation (e.g., always prioritizing care over authority), we would expect strongly asymmetric projections. Instead, both conflicting foundations contribute roughly equally to the within-subspace component, consistent with the model encoding the *conflict itself* rather than a pre-resolved moral judgment. This is an "understanding before choice" mode of representation: the model maintains both competing moral claims in tension rather than collapsing to a resolution, suggesting that moral comprehension (the capacity to represent the structure of ethical disagreement) may precede and be separable from moral judgment.

The complexity–fragility gradient (single-foundation $\sigma^* = 4.72 >$ pooled dilemma $\sigma^* = 3.12 >$ per-type dilemma $\sigma^* = 2.90$) shows that representational complexity trades off against robustness. Dilemma representations sit in a higher-dimensional space and are correspondingly more fragile under Gaussian noise. This is consistent with the residual component encoding subtle contextual features that require higher precision to maintain.

5.5 Register sensitivity: directions transfer, thresholds do not

Foundation-specific probes trained on declarative minimal pairs do not fully generalize to narrative dilemma text *as classifiers*. In the dilemma verification experiment, authority and loyalty probes showed near-chance transfer (Youden's $J < 0.2$), while care and fairness probes transferred well. This asymmetry is not a model-capacity issue: testing on OLMo-2 7B (32 layers, 4096 hidden dim) yielded comparable transfer failure (54.0% vs. 61.3% for the 1B model).

Directions vs. thresholds. The probe engineering analysis (§4.13) resolves this concern by separating two components of cross-register transfer: the *direction* (probe weight vector) and the *threshold* (bias term). When evaluated by pair accuracy (the fraction of pairs where the direction projects the moral text higher than the neutral text, requiring no threshold), both probe-weight and mean-difference directions achieve $> 90\%$ mean pair accuracy on narrative dilemma text (probe-weight $> 95\%$ for all foundations; mean-difference drops to $\sim$91% for authority/subversion). The Youden's J failure is therefore a threshold miscalibration effect: the absolute projection scale shifts between registers, invalidating the fixed decision boundary learned on declarative text. The directional structure itself, the subspace in which moral content is encoded, transfers robustly.

This distinction matters for the geometric findings. Cosine similarity, effective dimensionality, and dendrogram clustering depend only on direction vectors, not on classification thresholds. Since the directions transfer across registers, the geometric findings are not register-bound.

Connection to differential fragility. The register sensitivity pattern partially parallels the fragility findings (Section 5.2): binding foundations (authority, loyalty) that are most sensitive to text register are also among the most affected by MoE output dilution (sanctity in particular). However, the B.2 results show that this parallel is specific to threshold transfer, not direction transfer. Both individualizing and binding directions transfer with high pair accuracy ($> 90\%$), suggesting the differential fragility reflects genuine representational vulnerability to noise rather than register entanglement in the directions.

Implications for the geometric findings. The 21-direction dendrogram analysis (§4.9) gives direct evidence that register features drive part of the representation geometry: projecting all directions into the 5D moral subspace dissolves the categorical foundation/dilemma separation, confirming that the separation is carried by extra-moral (register) features. This is consistent with the threshold miscalibration account: foundation and dilemma directions occupy the same moral subspace (their projections overlap), but differ in extra-moral dimensions that carry register information and shift the activation scale.

5.6 Limitations

Small probing dataset. The 32 training pairs per foundation are sufficient for classification (near-perfect accuracy) but limit the precision of direction estimation. Bootstrap analysis (Section 4.6) confirms that directions at layers 0–4 are borderline unstable, and all geometric claims are qualified to layers 5–14. A larger dataset would tighten direction estimates, but the current dataset is deliberately minimal to demonstrate that structured geometry is recoverable even from small samples.

Permutation test power. With 6 foundations divided into two groups of 3, the permutation space contains only 20 unique partitions. The permutation test for the individualizing/binding distinction cannot reach $p < 0.05$ unless the effect is very large. The dendrogram structure gives qualitative support, but the formal statistical test is underpowered.

Two models, one lab. Both OLMo-2 and OLMoE are from Ai2 and trained on comparable corpora. The geometric findings may not generalize to models trained on substantively different data mixtures. The architectural comparison (dense vs. MoE) is well-controlled because the models share training data, but this comes at the cost of corpus diversity.

Linear probes. The entire analysis assumes that moral foundations are encoded as linear directions. If some foundations are encoded nonlinearly (e.g., as curved manifolds or distributed across multiple directions), the cosine similarity analysis would understate the true geometric richness. The near-perfect accuracy of linear probes suggests that linear decoding captures the dominant signal, but does not rule out additional nonlinear structure.

No causal evidence. Probe directions are correlational: they identify *where* foundation information is readable, not whether that information is *used* by the model during generation. Causal methods (activation patching, ablation studies) are needed to establish whether the geometric structure we observe is functionally relevant.

6 Conclusion

Language models develop structured moral representations that go beyond mere moral detection. By training independent linear probes for each of six Moral Foundations Theory foundations and analyzing the geometry of their weight vectors, we find that OLMo-2 1B and OLMoE-1B-7B exhibit the *integration* signature: foundation directions are distinct (effective dimensionality $= 5$), share a common moral-salience component (mean pairwise cosine ≈ 0.22–0.27), and span a high-dimensional moral subspace. This geometric structure is consistent across the two architectures tested (dense vs. MoE), emerges early in pre-training, and stabilizes before probing accuracy saturates. However, the inter-framework structure does not align with MFT's predicted individualizing/binding grouping, indicating that the model's moral taxonomy is empirically grounded in corpus statistics rather than organized along human-theoretical lines.

Framework-specific fragility testing shows that the output dilution effect in MoE models is not foundation-uniform: sanctity/degradation is the most robust foundation in the dense model but the most fragile in MoE (6.2$\times$ ratio), far exceeding the overall architectural gap.

Extending this geometric lens to moral dilemmas (scenarios where two foundations conflict) shows partial compositionality: dilemma representations share $\sim$10% of their variance with the subspace spanned by component foundation directions (100$\times$ the null baseline), with near-balanced loading across both conflicting foundations. The remaining $\sim$90% residual and the complexity–fragility gradient (σ^*: 4.72 $\rightarrow$ 3.12 $\rightarrow$ 2.90) indicate that dilemma representations encode conflict-specific structure beyond their component foundations. This compositionality pattern is preserved across architectures (dense and MoE).

The probe direction geometry methodology introduced here is general. Any set of related concepts that can be isolated by binary linear probes can be analyzed for inter-concept structure via the same cosine similarity and clustering techniques. We anticipate applications to other taxonomies of human values, to political orientation, and to the internal organization of safety training in instruction-tuned models.

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Appendices

Supplementary material.

A Per-Foundation Probe Accuracy Tables

A.1 OLMo-2 1B

Table 1: Per-foundation probe accuracy across layers, OLMo-2 1B. Each cell shows test accuracy (16 test examples per foundation). All foundations exceed 0.6 at every layer.

Layer	Care	Fairness	Liberty	Loyalty	Authority	Sanctity
0	1.000	1.000	1.000	0.938	1.000	0.938
1	0.875	0.938	0.938	0.938	1.000	0.938
2	0.938	0.938	0.938	0.938	1.000	0.938
3	0.938	0.938	0.938	0.938	1.000	0.938
4	1.000	0.938	1.000	0.938	1.000	0.938
5	1.000	0.938	1.000	1.000	1.000	1.000
6	1.000	0.938	1.000	1.000	1.000	0.938
7	1.000	1.000	1.000	1.000	1.000	0.938
8	1.000	1.000	1.000	1.000	1.000	1.000
9	1.000	0.938	1.000	1.000	1.000	0.938
10	1.000	0.938	1.000	1.000	1.000	0.938
11	1.000	0.938	1.000	0.938	1.000	0.938
12	1.000	0.938	1.000	0.938	1.000	1.000
13	1.000	0.938	1.000	1.000	1.000	1.000
14	1.000	0.938	1.000	1.000	1.000	0.938
15	1.000	0.938	1.000	0.938	1.000	0.938

A.2 OLMoE-1B-7B

Table 2: Per-foundation probe accuracy across layers, OLMoE-1B-7B.

Layer	Care	Fairness	Liberty	Loyalty	Authority	Sanctity
0	0.938	0.875	0.938	0.938	0.875	0.938
1	0.938	0.875	0.812	0.812	0.812	0.812
2	0.938	0.875	0.875	0.812	0.875	0.812
3	0.938	0.938	0.938	0.875	0.875	0.875
4	0.938	0.938	0.938	1.000	1.000	1.000
5	0.938	1.000	1.000	1.000	1.000	1.000
6	0.938	0.938	1.000	1.000	1.000	1.000
7	1.000	1.000	1.000	1.000	1.000	1.000
8	1.000	1.000	1.000	1.000	1.000	1.000
9	1.000	1.000	1.000	1.000	1.000	1.000
10	1.000	0.938	1.000	1.000	1.000	1.000
11	1.000	1.000	1.000	1.000	1.000	1.000
12	1.000	0.938	1.000	1.000	1.000	1.000
13	1.000	1.000	1.000	1.000	1.000	1.000
14	1.000	0.938	1.000	1.000	1.000	1.000
15	1.000	0.938	1.000	1.000	1.000	1.000

Table 3: Bootstrap direction stability (mean cosine similarity with full-data direction, 200 resamples). Values marked with * fall below the 0.8 stability threshold.

Layer	Care	Fairness	Liberty	Loyalty	Authority	Sanctity
0	0.740*	0.741*	0.768*	0.761*	0.780*	0.793*
1	0.752*	0.765*	0.775*	0.763*	0.780*	0.789*
2	0.774*	0.779*	0.796*	0.785*	0.790*	0.789*
3	0.767*	0.788*	0.791*	0.784*	0.808	0.811
4	0.789*	0.803	0.817	0.796*	0.812	0.819
5	0.799*	0.814	0.812	0.818	0.809	0.830
6	0.807	0.816	0.830	0.819	0.809	0.825
7	0.817	0.811	0.813	0.818	0.830	0.842
8	0.821	0.822	0.822	0.815	0.792*	0.807
9	0.821	0.816	0.832	0.814	0.820	0.803
10	0.823	0.809	0.831	0.821	0.817	0.836
11	0.817	0.802	0.830	0.837	0.833	0.814
12	0.828	0.801	0.831	0.843	0.828	0.820
13	0.825	0.811	0.821	0.817	0.827	0.838
14	0.821	0.823	0.824	0.842	0.826	0.827
15	0.816	0.826	0.803	0.847	0.838	0.852

B Bootstrap Direction Stability Tables

Layers 0–2 have widespread instability (< 0.8): all six foundations are unstable. Layers 3–5 are mixed (2–6 of 6 unstable). Layers 6–15 are mostly stable, with one exception (authority at layer 8, 0.792). Sanctity/degradation is the most stable foundation (13/16 layers above threshold); care/harm is the least stable (10/16).

The stable core (layers 6–15) gives the most reliable geometric measurements. The headline finding (integration signature, effective dimensionality = 5) is confirmed within this range.

C Cosine Similarity Matrices

Full pairwise cosine similarity matrices between the six foundation probe directions at selected layers of OLMo-2 1B.

Table 4: Cosine similarity matrix at layer 0 (peak separation), OLMo-2 1B. Mean off-diagonal = 0.216.

	Care	Fair	Lib	Loy	Auth	Sanc
Care	1.000	0.148	0.189	0.203	0.164	0.201
Fair	0.148	1.000	0.232	0.179	0.224	0.143
Lib	0.189	0.232	1.000	0.273	0.333	0.224
Loy	0.203	0.179	0.273	1.000	0.264	0.219
Auth	0.164	0.224	0.333	0.264	1.000	0.249
Sanc	0.201	0.143	0.224	0.219	0.249	1.000

Table 5: Cosine similarity matrix at layer 7, OLMo-2 1B. Mean off-diagonal = 0.262.

	Care	Fair	Lib	Loy	Auth	Sanc
Care	1.000	0.264	0.221	0.250	0.188	0.229
Fair	0.264	1.000	0.236	0.294	0.292	0.195
Lib	0.221	0.236	1.000	0.317	0.324	0.252
Loy	0.250	0.294	0.317	1.000	0.329	0.242
Auth	0.188	0.292	0.324	0.329	1.000	0.298
Sanc	0.229	0.195	0.252	0.242	0.298	1.000

Table 6: Cosine similarity matrix at layer 15, OLMo-2 1B. Mean off-diagonal = 0.262.

	Care	Fair	Lib	Loy	Auth	Sanc
Care	1.000	0.246	0.200	0.226	0.168	0.195
Fair	0.246	1.000	0.220	0.347	0.278	0.138
Lib	0.200	0.220	1.000	0.352	0.393	0.237
Loy	0.226	0.347	0.352	1.000	0.415	0.220
Auth	0.168	0.278	0.393	0.415	1.000	0.295
Sanc	0.195	0.138	0.237	0.220	0.295	1.000

At layer 0, the highest pairwise cosine similarity is liberty–authority (0.333), followed by liberty–loyalty (0.273). At later layers, the liberty–authority and loyalty–authority pairs consistently show the strongest similarity (0.393 and 0.415 at layer 15). These pairings cross the MFT individualizing/binding boundary, indicating that the model's inter-framework geometry is organized along empirical distributional axes rather than the theoretical MFT grouping.

D Permutation Tests for MFT Group Structure

We test whether the six foundation probe directions cluster into the MFT-predicted individualizing (care, fairness, liberty) and binding (loyalty, authority, sanctity) groups. The test statistic is the difference between mean within-group cosine similarity and mean between-group cosine similarity. We generate the null distribution by permuting group assignments 10,000 times.

Table 7: Permutation test for individualizing/binding group structure across layers, OLMo-2 1B. No layer reaches significance.

Layer	Observed statistic	p-value
0	0.001	0.394
1	−0.014	0.783
2	−0.003	0.696
3	−0.004	0.791
4	−0.004	0.788
5	0.003	0.438
6	−0.002	0.486
7	0.005	0.322
8	0.001	0.486
9	−0.003	0.582
10	−0.003	0.579
11	−0.014	0.777
12	−0.004	0.675
13	0.003	0.387
14	0.000	0.385
15	0.007	0.339

The test does not reach significance at any layer (minimum $p = 0.32$ at layer 7). The observed statistics are near zero and frequently negative (within-group similarity < between-group similarity), confirming that the model's inter-framework geometry is not organized along the MFT individualizing/binding axis. This is consistent with the dendrogram analysis (Section 4.3), which shows cross-MFT pairings (care–sanctity, liberty–authority) rather than the predicted group structure.

E Reproducibility

E.1 Hardware and software

All experiments ran on a single MacBook Pro with Apple M4 Pro (24 GB unified memory) using the MPS backend. Software versions: Python 3.13, PyTorch 2.7, Transformers 4.49.

E.2 Runtime

Experiment	Model	Wall time
1–3 (probing + geometry + bootstrap)	OLMo-2 1B	$\sim$5 min
5 (dense vs. MoE geometry)	OLMoE-1B-7B	$\sim$20 min
6 (geometric trajectory)	OLMo-2 1B (20 ckpts)	$\sim$45 min
7 (framework fragility)	OLMo-2 + OLMoE	$\sim$25 min

E.3 Reproducibility notes

Probe training. Linear probes are `nn.Linear(2048, 1)` trained with BCE loss and Adam (lr = 10^{-2}) for 50 epochs. No weight decay, no learning rate schedule. Random seed is not fixed across runs; bootstrap analysis (Appendix~B) quantifies direction stability under resampling.

Probing dataset. The 240-pair minimal-pair dataset (40 per MFT foundation) is deterministic and version-controlled. Dataset generation uses Claude Sonnet 4.6 with automated validation gates (embedding similarity, keyword scan, LLM-as-judge filtering); the exact dataset is included in the code repository.

Activation collection. Forward passes use `torch.no_grad()`. Activations are mean-pooled across the sequence dimension at each layer. For OLMoE, activations are collected after the expert combination step (post-routing), not from individual experts.

Code availability. All experiment scripts, the probing dataset, and figure generation code are available at `https://github.com/deepsteer/deepsteer`.